

Manasi Somasundaram

Research Data Analyst

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SUMMARY

Experienced **Research Data Analyst** with over 4+ years supporting **clinical research** and **healthcare studies** in academic, hospital, and life sciences settings. Skilled in managing and analysing complex patient data to support **real-world evidence (RWE)**, clinical trials, and regulatory reporting. Adept at ensuring compliance with **HIPAA**, **GCP**, and **FDA** guidelines while collaborating with physicians, coordinators, and research teams. Proven track record in improving **data accuracy**, streamlining workflows, and supporting multi-disciplinary studies across **oncology**, **cardiology**, and **infectious diseases**. Known for strong attention to detail, problem-solving, and cross-functional communication to drive **data-driven insights** in clinical research environments.

TECHNICAL SKILLS

Clinical Trials: Protocols, CRF, AE/SAE Reporting, MedDRA, CTCAE, eTMF, SDV, FDA IND, ICH-GCP

EDC Systems: REDCap, Medidata Rave, Oracle Clinical, Veeva Vault

Statistical Tools: R, Python, SPSS, SAS; T-test, Chi-square, ANOVA, Regression, Survival Analysis

Visualisation: Tableau, Power BI, Excel (Pivot Tables, Charts), Jupyter

EHR & Standards: EPIC, eClinicalWorks; ICD-10, HL7, FHIR, DICOM

Data Management: SQL, ETL, RWD, Genomic Data, Data Cleaning, Query Resolution

Compliance & QA: HIPAA, 21 CFR Part 11, ICH-GDP, PHI, SOPs, Audit Readiness

Collaboration Tools: Jira, Confluence, GitHub, SharePoint, Teams

Soft Skills: Teamwork, Communication, Documentation, Detail-Oriented, Deadline-Driven

CERTIFICATIONS

AHIMA, RHIA Certification - Current

LinkedIn Learning: R language - May 2023

Coursera: Python Programming - Jan 2022

HIPAA Certification - Jan 2022

PROFESSIONAL EXPERIENCE

Elevance Health, USA

Feb 2024 - Current

Research Data Analyst

- Collect, validate, and manage **clinical and demographic data** from diverse patient populations using **Epic EHR**, ensuring accuracy for multi-disciplinary research studies across cardiology, oncology, and infectious disease.
- Coordinate with **IRB teams** and **clinical investigators** to ensure data usage complies with **HIPAA**, **PHI** guidelines, and **21 CFR Part 11** standards; contribute to audit trails and documentation for FDA and NIH-funded projects.
- Extract structured and unstructured data from **EHR** and **REDCap** databases using **SQL**, **Python** (Pandas, NumPy), and **R**, optimizing data pipelines for analysis and reporting.
- Design and automate visual dashboards using **Power BI** and **Excel** to track **patient enrollment**, **adverse events**, and study progress across multiple **clinical trials**, reducing manual reporting time by 30%.
- Participate in data cleaning, transformation, and harmonisation of real-world data (RWD), including patient-reported outcomes and laboratory results, to ensure completeness and readiness for statistical analysis.
- Communicate with physicians, coordinators, and biostatisticians to resolve discrepancies in medical records, adverse event logs, and case report forms (CRFs), improving data quality and protocol adherence.
- Maintain study documentation, workflows, and version-controlled datasets using SharePoint, Jira, and Confluence, supporting data traceability and collaboration among research teams.

Indiana University School of Medicine – Hands in Autism Center

May 2023 - Dec 2023

Clinical Data Analyst Intern

- Cleaned and validated **behavioural health datasets** using Python and R to improve data accuracy for research purposes.
- Created dashboards and visual reports in **Tableau** and **Power BI** to summarise patient progress and treatment outcomes.
- Applied **ICD-10** coding and **HL7/FHIR** standards to integrate clinical data from multiple sources.
- Assisted in preparing study documentation and reports following **Good Clinical Practice - GCP** guidelines.

CitiusTech, India

Jun 2020 - Dec 2021

Clinical Data Analyst, Oncology Department

- Accurately entered and updated oncology clinical trial data into **Medidata Rave** and **Oracle Clinical**, adhering to study protocols and timelines.
- Conducted detailed reviews of adverse event (AE) and serious adverse event (SAE) reports, applying **MedDRA** coding and **CTCAE** grading to ensure proper classification and compliance with regulatory requirements.
- Performed source data verification (**SDV**) by cross-checking clinical trial data against source documents to identify and correct discrepancies.
- Responded to **data queries** raised by clinical monitors and project teams, facilitating timely resolution to maintain data quality and trial integrity.
- Managed electronic Trial Master File (**eTMF**) in **Veeva Vault**, organising essential study documents and tracking updates to support audit and inspection readiness.
- Collaborated with cross-functional teams, including clinical operations and data management, to support **data cleaning**, **study closeout** activities, and adherence to Good Clinical Practice (**GCP**) guidelines.

Research Data Analyst

- Extracted large volumes of **genomic and clinical data** from multiple databases using advanced **SQL** queries, ensuring accurate and timely data retrieval for ongoing research projects.
- Cleaned, transformed, and validated raw datasets with **Python** libraries such as **Pandas** and **NumPy** to improve data quality and reliability for downstream analysis.
- Designed and implemented **ETL** workflows to automate the integration and standardisation of heterogeneous data sources, reducing manual processing time by 30%.
- Performed **statistical analyses**, including regression models and survival analysis using **Python** and **R** to identify significant biomarkers and patient outcome trends in oncology studies.
- Created detailed data reports and interactive **visualisations** with **Tableau** and **Matplotlib** to effectively communicate insights to clinical researchers and project stakeholders.

EDUCATION

Master of Science in Health Informatics (Dec 2023)

Indiana University, Indianapolis, USA

Doctor of Pharmacy (Aug 2021)

Osmania University, India

PROJECTS

Genomic Data Analysis of Multiple Myeloma Patient Samples

- Conducted large-scale genomic data analysis on 1,300+ patient samples using **R**, identifying **high-impact mutations** to support precision oncology initiatives.
- Designed and implemented ETL pipelines to integrate genomic datasets with EHR systems, streamlining clinical data flow and improving decision-making for treatment strategies.

Lung Cancer Data Analysis and Visualisation

- Executed data analysis on a **10,000+ record lung cancer dataset** using **SQL**; utilized **Tableau** to build dynamic dashboards for real-time insights.
- Discovered and visualized **smoking** as a primary risk factor through **data correlation** and pattern analysis, supporting evidence-based public health interventions.